Macroeconomics I — MPE Insper

Practice list: Measurement, Solow and Growth Accounting (Listas 1–2)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Measuring output, prices and welfare *(Kurlat chs. 1–2 • 25 points)*

Question 1. (10 points) Objective items. IBGE's national accounts team tests its procedures on a two-good economy (food, good 1; electronics, good 2):

	p_1	q_1	p_2	q_2
2024	2	10	10	5
2025	3	12	7	8

a) (3 points) With 2024 as the base year, the GDP deflator for 2025 (2024 = 100) is:

A. 92.86

B. 131.43

C. 88.46

D. 113.04

Answer: C

Step — nominal and real. Nominal GDP 2025 $= 3 \cdot 12 + 7 \cdot 8 = 92$. Real GDP 2025 at 2024 prices $= 2 \cdot 12 + 10 \cdot 8 = 104$. Deflator $= 100 \cdot 92/104 = 88.46$.

Step — distractors. 92.86 $= 65/70$ is the Laspeyres (fixed 2024 basket) price index, not the deflator; 131.43 $= 92/70$ is nominal GDP growth; 113.04 inverts the ratio.

Ref: Kurlat (2020), §1.2, p. 22.

b) (3 points) Real GDP growth between 2024 and 2025 computed with Kurlat's chained method (geometric average of the growth rates at initial-year and final-year prices) is:

A. 48.57%

B. 31.43%

C. 41.54%

D. 45.01%

Answer: D

Step — two growth rates. At 2024 prices: $1 + g^I = 104/70 = 1.4857$. At 2025 prices: real 2024 $= 3 \cdot 10 + 7 \cdot 5 = 65$, so $1 + g^F = 92/65 = 1.4154$.

Step — chain. $1 + g = (1.4857 \cdot 1.4154)^{1/2} = 1.4501$, i.e. 45.01%. Option B is nominal growth.

Economic intuition. Electronics grows most and its relative price falls, so early-year prices weight the fast-growing good more and give the higher rate; chaining splits the difference.

Ref: Kurlat (2020), §1.2, "Alternative 3: chained prices", pp. 23–24.

c) (2 points, T or F) If the IPCA, a Laspeyres index with a fixed basket, rises 6% in a year, then the extra income a household needs to reach its original utility level rose by *at least* 6%.

Answer: F

Step — substitution bias. The fixed basket still affords the old utility, so 6% is enough; the household can do better by substituting toward goods that became relatively cheaper. The Laspeyres index is an *upper* bound on the true cost-of-living change: at most 6%.

Ref: Kurlat (2020), §1.2; formula sheet `rules/01_mensuracao_agregados.md`.

d) (2 points, T or F) Suppose tradable goods obey the law of one price at the market exchange rate and non-tradable services are cheaper in Brazil than in the US. Then converting Brazil's GDP to dollars at the market exchange rate understates it relative to a PPP conversion.
Answer: T

Step — why. At the market rate tradables are priced the same in both countries, but Brazil's non-tradables are valued at their (low) Brazilian price. PPP values them at common prices, so Brazilian output is worth more at PPP (Balassa–Samuelson).

Ref: Kurlat (2020), §1.2, "Comparisons across countries and PPP".

Question 2. (15 points) Discursive. A Penn World Table analyst compares Brazil and the US. Each produces a tradable good T and a non-tradable service N. Prices are in local currency; the market exchange rate is $e = 5$ reais per dollar.

	p_T	q_T	p_N	q_N
Brazil (reais)	10	50	20	100
US (dollars)	2	200	20	150

a) (5 points) Compute Brazil's GDP in dollars (i) at the market exchange rate and (ii) at PPP, using US prices as the common prices. Compute Brazil/US GDP under each measure and the implied PPP exchange rate. Explain the gap.

Step — local GDP. Brazil: $10 \cdot 50 + 20 \cdot 100 = 2500$ reais. US: $2 \cdot 200 + 20 \cdot 150 = 3400$ dollars.

Step — market rate. $2500/5 = 500$ dollars; Brazil/US $= 500/3400 = 0.147$.

Step — PPP. Brazil's quantities at US prices: $2 \cdot 50 + 20 \cdot 100 = 2100$ dollars; Brazil/US $= 2100/3400 = 0.618$. Implied PPP rate: $2500/2100 = 1.19$ reais per dollar.

Step — gap. Tradables obey the law of one price ($10/2 = 5 = e$), but the service costs $20/5 = 4$ dollars in Brazil against 20 in the US. The market rate prices Brazil's large service sector at a fifth of its PPP value, so Brazil looks 4.2 times poorer than it is.

Economic intuition. Exchange rates are set by traded goods; living standards depend on all goods. Poorer countries have cheap non-tradables (low wages in services), so the market rate understates them.

Ref: Kurlat (2020), §1.2, pp. 24–27.

Grading rubric.

- Local GDP of both countries: 1 pt.
- Market-rate conversion and ratio: 1 pt.
- PPP value, ratio and implied PPP rate: 2 pts.
- Explanation through non-tradables / law of one price: 1 pt. Saying only "PPP is better" with no mechanism: 0 for this line.
- Converting PPP by dividing by e again (double conversion): max. 2 pts in the item.
- Arithmetic slip with correct method: -1 pt.

b) (5 points) Now use the Jones–Klenow welfare measure of Kurlat §2.2 with $\sigma = 1$ and equal leisure in both countries, so that a random person's utility is $\mathbb{E}[a(\bar{u} + \ln c)]$, where $a = 1$ if alive. Normalize $c_{US} = 1$, set $\bar{u} = 5$, survival probabilities $a_{US} = 0.79$, $a_{BR} = 0.76$, and Brazilian consumption per person $\bar{c}_{BR} = 0.30$, *equal for everyone*. Derive λ (the scaling of US consumption that leaves Rawls indifferent), compute it, and split the welfare gap into a consumption part and a life-expectancy part.

Step — indifference. $a_{US}\,[\bar{u} + \ln(\lambda c_{US})] = a_{BR}\,[\bar{u} + \ln \bar{c}_{BR}]$, so $\ln \lambda = \dfrac{a_{BR}}{a_{US}}(\bar{u} + \ln \bar{c}_{BR}) - \bar{u}$.

Step — number. $\ln\lambda = \frac{0.76}{0.79}(5 - 1.204) - 5 = -1.348$, so $\lambda = 0.260$.

Step — decomposition. With equal survival λ would be 0.30. Lower life expectancy multiplies it by $0.260/0.30 = 0.866$: about 13% of consumption-equivalent welfare.

Economic intuition. $\bar{u}$ is the value of being alive. Three fewer years of life expectancy cost Rawls the chance of enjoying $\bar{u} + \ln c$, so he needs less US consumption to be indifferent.

Ref: Kurlat (2020), §2.2, eqs. (2.2.1)–(2.2.2), pp. 33–41.

Grading rubric.

- Writes the indifference condition with a multiplying the whole bracket: 2 pts.
- Solves for $\ln\lambda$ and computes $\lambda \approx 0.26$: 2 pts.
- Decomposition and interpretation: 1 pt.
- Multiplying only $\ln c$ by a (forgetting $\bar{u}$): max. 2 pts in the item.

c) (5 points) Now half of Brazilians consume 0.12 and half 0.48 (same mean 0.30). Recompute λ. By how much does inequality lower it? Would a higher σ make the penalty larger or smaller? Why does Brazil fall further behind in welfare than in consumption?

Step — expected log. $\mathbb{E}\ln c = \frac{1}{2}\ln 0.12 + \frac{1}{2}\ln 0.48 = \ln\sqrt{0.0576} = \ln 0.24$.

Step — new λ. $\ln\lambda = \frac{0.76}{0.79}(5 + \ln 0.24) - 5 = -1.563$, so $\lambda = 0.210$.

Step — penalty. $0.210/0.260 = 0.807$: inequality removes about 19%. It acts through the ratio geometric/arithmetic mean $0.24/0.30 = 0.8$, scaled by survival ($0.8^{0.96}$).

Step — role of σ. σ is the curvature of u. With $\sigma > 1$, utility is more concave and the poor half's low consumption weighs more: the penalty grows.

Economic intuition. Behind the veil of ignorance, inequality is risk. A risk-averse Rawls pays to avoid it, so an unequal country with the same mean consumption is worth less to him.

Ref: Kurlat (2020), §2.2, risk aversion and eq. (2.2.5), pp. 35–38.

Grading rubric.

- Uses $\mathbb{E}\ln c$, not $\ln\mathbb{E}c$: 2 pts. Using $\ln 0.30$: max. 1 pt in the item.
- New $\lambda \approx 0.21$ and the penalty: 2 pts.
- Direction of σ with the concavity argument: 1 pt.

Block 2 — Solow, the Golden Rule and growth accounting *(Kurlat chs. 4–5 • 25 points)*

Question 3. (10 points) Objective items. IPEA calibrates a Solow model *per worker*, without technological progress: $y = k^{\alpha}$, $\alpha = 1/3$, $s = 0.20$, $n = 0.02$, $\delta = 0.08$.

a) (3 points) Steady-state output per worker y^* is:

A. 2.828

B. 1.131

C. 1.414

D. 2.000

Answer: C

Step — steady state. $k^* = \left(s/(n+\delta)\right)^{1/(1-\alpha)} = 2^{3/2} = 2.828$; $y^* = (k^*)^{1/3} = 2^{1/2} = 1.414$. Option A is k^*; option B is $c^* = (1-s)y^*$.

Ref: Kurlat (2020), §4.2, p. 57.

b) (3 points) Moving the economy to the Golden Rule saving rate raises steady-state consumption per worker by approximately:

A. 7.6%

B. 0%, since $s = 0.20$ already maximizes c^*

C. 33.3%

D. 16.7%

Answer: A

Step — Golden Rule. $f'(k_{gr}) = n + \delta$ gives $s_{gr} = \alpha = 1/3$ and $k_{gr} = (\alpha/(n+\delta))^{3/2} = 6.086$, $c_{gr} = 6.086^{1/3} - 0.1 \cdot 6.086 = 1.217$.

Step — gain. $c^* = 0.8 \cdot 1.414 = 1.131$; $1.217/1.131 - 1 = 7.6\%$.

Ref: Kurlat (2020), §4.3, p. 61.

c) (2 points, T or F) Because this economy is below the Golden Rule, raising s to s_{gr} is a Pareto improvement.

Answer: F

Step — transition. Below the Golden Rule, reaching k_{gr} requires *lower* consumption during the transition. Current generations lose; only the reverse case ($k^* > k_{gr}$, dynamic inefficiency) gives a free lunch.

Ref: Kurlat (2020), §4.3.

d) (2 points, T or F) If $g_Y = 3\%$, $g_K = 4.5\%$, $g_L = 1.5\%$ and $\alpha = 1/3$, then all growth of output per worker comes from capital deepening.

Answer: F

Step — accounting. $g_y = 3 - 1.5 = 1.5\%$; capital deepening $\alpha g_k = \frac{1}{3}(4.5 - 1.5) = 1.0\%$; residual $g_A = 0.5\%$. A third of g_y is TFP.

Ref: Kurlat (2020), §5.4, eq. (5.4.2), p. 86.

Question 4. (15 points) Discursive. Brazil is modeled as a Solow economy with labor-augmenting technology: $Y = K^\alpha (AL)^{1-\alpha}$, $\alpha = 1/3$, $s = 0.24$, $n = 0.01$, $g = 0.02$ (growth of A), $\delta = 0.05$. Work *per efficiency unit*: $\tilde{k} = K/(AL)$, $\tilde{y} = Y/(AL)$.

a) (5 points) Derive the law of motion of $\tilde{k}$ (continuous time) and compute $\tilde{k}^*$, $\tilde{y}^*$, $\tilde{c}^*$, K/Y and the marginal product of capital in steady state. Give the growth rates of $\tilde{y}$, Y/L and Y on the balanced growth path. Is Brazil above or below the Golden Rule? Prove it with the marginal product, then confirm with $\tilde{c}$.

Step — law of motion. $\dot{\tilde{k}}/\tilde{k} = \dot{K}/K - g - n$ and $\dot{K} = sY - \delta K$, so $\dot{\tilde{k}} = s\tilde{k}^\alpha - (n + g + \delta)\tilde{k}$.

Step — steady state. $\tilde{k}^* = (0.24/0.08)^{3/2} = 3^{3/2} = 5.196$, $\tilde{y}^* = 3^{1/2} = 1.732$, $\tilde{c}^* = 0.76 \cdot 1.732 = 1.316$, $K/Y = \tilde{k}^*/\tilde{y}^* = s/(n+g+\delta) = 3$.

Step — growth. $\tilde{y}$: 0; Y/L: $g = 2\%$; Y: $n + g = 3\%$.

Step — Golden Rule test. $f'(\tilde{k}^*) = \alpha\tilde{y}^*/\tilde{k}^* = \frac{1}{3} \cdot \frac{1}{3} = 0.111 > n + g + \delta = 0.08$. Capital is scarce relative to the Golden Rule: *below* it. Check: $s_{gr} = 1/3$ gives $\tilde{k}_{gr} = (0.333/0.08)^{3/2} = 8.505$ and $\tilde{c}_{gr} = 1.361 > 1.316$.

Economic intuition. The return on capital net of what it costs to maintain $\tilde{k}$ ($n + g + \delta$) is positive, so more capital would raise steady-state consumption.

Ref: Kurlat (2020), §4.3 and §4.5, pp. 61–72.

Grading rubric.

- Law of motion with $(n + g + \delta)$: 1 pt. Omitting g: max. 2 pts in the item.
- Steady-state values and K/Y: 2 pts.
- BGP growth rates: 1 pt. Saying $\tilde{y}$ grows at g: -1 pt.
- Golden Rule comparison with a correct criterion: 1 pt.

b) (5 points) Floods destroy 30% of the capital stock at t_0 (illustrative). Using the log-linear approximation $\ln \dot{\tilde{y}} \approx -\lambda(\ln \tilde{y} - \ln \tilde{y}^*)$, derive λ, compute the half-life of the gap, the impact effect on Y/L and its growth rate just after the shock. Separately, suppose emigration permanently lowers n to 0.005: compute the change in $\tilde{y}^*$ and the long-run growth of Y/L and of Y.

Step — speed. Linearize $\dot{\tilde{k}}/\tilde{k} = s\tilde{k}^{\alpha-1} - (n+g+\delta)$ in $\ln \tilde{k}$ at the steady state: derivative $(\alpha-1)s\tilde{k}^{*\alpha-1} = -(1-\alpha)(n+g+\delta)$. Since $\ln \tilde{y} = \alpha \ln \tilde{k}$, the same speed applies: $\lambda = (1-\alpha)(n+g+\delta) = \frac{2}{3} \cdot 0.08 = 0.0533$.

Step — half-life. $\ln 2/\lambda = 13.0$ years.

Step — impact. L and A do not jump, so Y/L falls by $1 - 0.7^{1/3} = 11.2\%$.

Step — growth after. $\ln \tilde{y} - \ln \tilde{y}^* = \frac{1}{3} \ln 0.7 = -0.119$; $\tilde{y}$ grows at $0.0533 \cdot 0.119 = 0.63\%$, so Y/L grows at $2 + 0.63 = 2.63\%$ and decelerates toward 2%.

Step — lower n. $\tilde{y}^* = (0.24/0.075)^{1/2} = 1.789$: +3.3% level effect. Long-run growth of Y/L stays $g = 2\%$; growth of Y falls from 3% to 2.5%.

Economic intuition. Destroyed capital raises its marginal product, so recovery is fast at first. Demography changes the level of $\tilde{y}^*$ and the growth of the aggregate, never the growth of income per person, which is pinned by g.

Ref: Kurlat (2020), §4.2 and §4.5; Romer (2012), §1.5 (speed of convergence).

Grading rubric.

- Derives $\lambda = (1-\alpha)(n+g+\delta)$: 1.5 pts. Quoting it without derivation: 0.5 pt.
- Half-life and impact effect: 1.5 pts.
- Growth just after the shock: 1 pt.
- Effect of lower n, both growth rates: 1 pt. Saying long-run growth of Y/L rises: max. 3 pts in the item.

c) (5 points) Development accounting with $y = A\,k^{\alpha}h^{1-\alpha}$. Brazil/US ratios: $y = 0.25$, $k = 0.20$, $h = 0.60$. Compute the TFP ratio and the share of the log income gap due to k, h and TFP. Then suppose a quarter of Brazil's measured capital is security capital (walls, armored cars) that produces nothing. Recompute the TFP ratio on productive capital and explain what the residual measured in the first step was capturing.

Step — TFP. $A_{BR}/A_{US} = 0.25/(0.2^{1/3} \cdot 0.6^{2/3}) = 0.25/(0.585 \cdot 0.711) = 0.601$.

Step — shares. $\ln 0.25 = -1.386$. Capital: $\frac{1}{3} \ln 0.2 / \ln 0.25 = 38.7\%$; human capital: $\frac{2}{3} \ln 0.6 / \ln 0.25 = 24.6\%$; TFP: $\ln 0.601 / \ln 0.25 = 36.7\%$. They add to 100%.

Step — security capital. Productive k ratio $= 0.75 \cdot 0.2 = 0.15$: $A_{BR}/A_{US} = 0.25/(0.15^{1/3} \cdot 0.711) = 0.661$.

Step — interpretation. Counting security capital as productive overstates Brazil's inputs, so the measured residual (0.601) is lower than the TFP of productive capital (0.661). The residual captures crime as if it were bad technology: it bundles technology with misallocation and institutions.

Economic intuition. TFP is "a measure of our ignorance": whatever inputs do not explain lands in it, including policies and resource allocation.

Ref: Kurlat (2020), §5.3–5.5, pp. 80–95.

Grading rubric.

- TFP ratio: 1.5 pts.
- Log decomposition adding to one: 1.5 pts.
- Recomputed TFP with productive capital: 1 pt.
- Interpretation of the residual: 1 pt. Calling the residual "pure technology": 0 for this line.